Base-level one-hot 1D convolutional networks for high-cardinality viral taxonomic classification: accuracy, corruption trade-offs, and learned sequence representations

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Abstract

Background: High-cardinality viral taxonomic classification requires models capable of distinguishing closely related genomic sequences while maintaining predictive performance as sequence quality deteriorates. We investigated base-level one-hot DNA representations combined with 1D adaptations of ResNet50 and ConvNeXt in a closed-set taxonomic classification setting comprising 13,683 viral species and 2,702 genera. Models were evaluated on held-out 512-bp sequence fragments.

[**Nota v2:** [Coerência]

Methods: We assessed robustness under controlled nucleotide resampling with effective change rate $3\epsilon/4$, the effect of noise-aware fine-tuning ($\epsilon = 0.05$), the organization of latent representations via cosine KNN ($k = 5$) and Silhouette scores, and the predictive contribution of first-layer sequence patterns through matched GC/complexity perturbations. Sliding-window consensus inference (512 bp window, 256 bp stride) was audited on long genomic contigs from NCBI.

[**Nota v2:** [Contribuição]

Results: On uncorrupted species-level fragments, standard ResNet50 achieved the highest accuracy (84.88%, Macro-F1 84.12%), but decreased rapidly under nucleotide corruption. At $\epsilon = 0.15$ (expected effective mutation rate of 11.25%), noise-aware ConvNeXt retained 17.38% accuracy compared with 3.87% for noise-aware ResNet50 and 0.30% for standard ResNet50. Species-level pairwise analysis at $\epsilon = 0.10$ showed ConvNeXt superior in 11,826 species, ResNet50 in 1,202, and 5 exact ties across 13,033 evaluated species. Analysis of first-layer filters identified GC-rich and homopolymeric patterns acting primarily as compositional detectors. On long NCBI contigs, standard ResNet50 achieved 72.69% accuracy on seen reference accessions, while ConvNeXt dropped to 3.49%–20.38% due to length bias in simulation and logit pulling in pre-softmax averaging.

[**Nota v2:** Pendências editoriais gerais. (i) Confirmar se o Ulisses deseja incluir o grupo (Microbial Systems Data Science) na afiliação. (ii) Periódicos BMC, incluindo a BioData Mining, exigem resumo estruturado (Background, Results, Conclusions); os três parágrafos do resumo já seguem essa lógica e podem receber os subtıtulos. Na v2, a seção inicial foi renomeada para Background e os Métodos foram movidos para antes dos Resultados. (iii) Sugere-se moderar o tıtulo: "enable robust" não é sustentado pelos resultados (acurácia de 3,87–17,38% em $\epsilon = 0{,}15$ e o próprio resumo descreve um trade-off). Exemplo: "Base-level one-hot 1D convolutional networks for high-cardinality viral taxonomic classification: accuracy, robustness, and learned representations". (iv) Todos os resultados derivam de uma única execução de treinamento por configuração; recomenda-se reportar intervalos de confiança (p.ex., bootstrap sobre o conjunto de teste) e, idealmente, variabilidade entre sementes.]

Conclusions: The evaluated 1D convolutional architectures exhibit a pronounced trade-off between clean-sequence discriminative power and corruption robustness. Rather than indicating unconstrained general robustness or functional sequence motifs, base-level one-hot 1D CNNs provide an effective, interpretable closed-set framework when controlled for base composition, reference overlap, and calibration.

Keywords: Viral Taxonomic Classification; Deep Learning; 1D Convolutional Neural Networks; Genomic Sequence Representation; Sequence Corruption; Robustness; Latent Representations; Sequence Pattern Perturbation.

1 Background

Advances in high-throughput DNA sequencing have enabled genomic data to be generated at increasingly large scales and across a broad range of short- and long-read technologies. In metagenomic analysis, converting these sequence reads into reliable taxonomic assignments remains a major computational challenge, particularly as reference databases increase in size and taxonomic resolution.

[**Nota v2:** [Ref.]

Widely used alignment-free methods address this problem by matching sequences against reference databases. Kraken 2, for example, uses exact k-mer matches together with lowest-common-ancestor assignment to achieve efficient taxonomic classification. Exact nucleotide matching, however, can be disrupted by sequence errors or other local perturbations. Alignment-based approaches such as BLAST provide a different trade-off between sensitivity and computational cost.

[**Nota v2:** [Contribuição]

Deep-learning approaches provide an alternative in which discriminative features are learned directly from nucleotide sequences. Neural taxonomic classifiers have been applied to metagenomic and viral classification problems using both flat and hierarchical prediction strategies, including GeNet (Rojas-Carulla et al., 2019), DeepMicrobes, CHEER, and MetageNN. GeNet pioneered residual networks on one-hot metagenomic reads; the present work extends this frontier by scaling closed-set taxonomic classification to 13,683 viral species and 2,702 genera, establishing controlled corruption trade-offs with corrected effective mutation rates ($3\epsilon/4$), formal calibration metrics (ECE, Brier, NLL), paired compositional motif controls, and windowed long-sequence voting mechanics.

[**Nota v2:** [Contribuição]

[**Nota v2:** Seção de Métodos movida para antes dos Resultados na v2, seguindo o formato usual dos periódicos BMC (Background, Methods, Results, Discussion, Conclusions) e porque os Resultados dependem de definições introduzidas aqui (ϵ e taxa efetiva de troca, nomenclatura dos modelos, protocolo KNN, critérios de validação dos motivos e mecanismo Voter). Confirmar a ordem exigida nas instruções da BioData Mining.]

2 Methods

2.1 Taxonomic Curation Pipeline and Hierarchy

Taxonomic reference labels were curated combining the International Committee on Taxonomy of Viruses (ICTV) Master Species List 2025 (MSL41) and the NCBI Taxonomy database accessed in 2025. The curated taxonomy comprises 13,683 Species, 2,702 Genera, and 247 Families.

[**Nota v2:** [Ref.]

[**Nota v2:** Informar: número de fragmentos por classe e total (o conjunto de teste tem 1,706,270 fragmentos, o que sugere ≈17 milhões no total); como os fragmentos de 512 bp foram gerados a partir dos genomas (com ou sem sobreposição, passo utilizado); e quantos genomas de origem há por espécie. Esses dados são necessários para avaliar o risco de fragmentos sobrepostos do mesmo genoma aparecerem em partições diferentes.]

2.2 Dataset Partitioning and Split Audit

Sequences were segmented into 512-bp non-overlapping fragments and partitioned according to an 80/10/10 stratified split across classes using seed 42. While no identical 512-bp sequence fragment appears across partitions, audit results (AD-007, S02/S03) establish that fragments derived from the same underlying viral genome accessions are present across train, validation, and test sets. Consequently, evaluation metrics measure closed-set fragment classification rather than generalization to unseen viral genomes.

[**Nota v2:** A ausência de sequências idênticas não exclui fragmentos parcialmente sobrepostos ou quase idênticos (do mesmo genoma ou de genomas muito similares) entre treino e teste, o que pode inflar a acurácia. Esclarecer e, se possıvel, avaliar também uma partição por genoma de origem.]

[**Nota v2:** A menção a L = 1024 bp foi removida, pois nenhum modelo com essa janela é avaliado no artigo.]

2.3 Long Sequence Dataset and Voter Mechanism

Variable-length contigs and complete genomes were retrieved from NCBI GenBank. Sequences were classified using a sliding window of length $W = 512$ bp and stride $S = 256$ bp. Window logits were aggregated by pre-softmax averaging:

$$\bar{z} = \frac{1}{N_w} \sum_{i=1}^{N_w} z_i, \quad \hat{y} = \arg\max \bar{z}. \tag{1}$$

Contigs shorter than 512 bp ($M < W$) were handled via end-padding with null vectors (`handle_short='pad'`), ensuring consistent evaluation across all contig lengths.

[**Nota v2:** Especificar os critérios de inclusão, o número de sequências por espécie, a distribuição de comprimentos e a data de acesso.]

[**Nota v2:** Especificar quais modelos (ou fases) foram treinados em cada GPU; a troca de hardware durante o treinamento afeta a reprodutibilidade numérica.]

[**Nota v2:** Informar se o stem do ResNet50 inclui BatchNorm, ReLU e MaxPool (como descrito para o ConvNeXt), as larguras de canal de cada estágio e a dimensão do vetor latente antes da camada final.]

[**Nota v2:** [Coerência]

[**Nota v2:** Informar como foram tratadas sequências com M < W e o critério de desempate da votação majoritária.]

2.4 Model Architectures and Spatial Nucleotide Encoding

Each sequence fragment was encoded as a 4-channel binary spatial tensor corresponding to nucleotides A, C, G, and T at channels 0, 1, 2, and 3, respectively. Ambiguous nucleotides and padding positions were encoded as all-zero null vectors. Two 1D convolutional neural network architectures were evaluated:

- **1D-ResNet50:** Adapted with a 1D convolutional stem (kernel size 7, stride 2, 64 channels) followed by BatchNorm, ReLU, and MaxPool, and four residual stages (channel widths 256, 512, 1024, 2048) yielding a 2048-dimensional latent vector before the final classification head.
- **1D-ConvNeXt:** Adapted with a 1D stem (kernel size 7, stride 4, 96 channels) followed by LayerNorm1d and four stages of depthwise separable blocks (widths 96, 192, 384, 768) with 768-dimensional latent representation.

[**Nota v2:** Informar o tamanho de lote do treinamento principal, o critério de seleção do checkpoint final (p.ex., menor perda de validação) e se o backbone permaneceu congelado durante o warmup dos modelos treinados do zero. Notar que os orçamentos de treinamento diferem entre modelos (p.ex., 154 épocas para o ResNet50 de espécie, 95 para o ConvNeXt do zero e 117 para o ConvNeXt por transferência, além das 133 épocas do modelo de gênero), o que é um fator de confusão na comparação entre arquiteturas e deve constar nas Limitações.]

[**Nota v2:** [Ref.]]

[**Nota v2:** [Ref.]]

2.5 Controlled Nucleotide Resampling Model

Controlled sequence corruption was simulated via a Bernoulli process: each position was selected with probability $\epsilon \in \{0, 0.02, 0.05, 0.10, 0.15\}$ and resampled uniformly from $\{A, C, G, T\}$, including the original base. The expected effective substitution rate is:

$$r_{\text{effective}} = \frac{3}{4}\epsilon. \tag{2}$$

At $\epsilon = 0.15$, the effective nucleotide corruption rate is exactly 11.25%.

[**Nota v2:** [Ref.]]

[**Nota v2:** [Ref.]]

2.6 Optimization and Training Schedules

Models were optimized using AdamW (Loshchilov and Hutter, 2019) with cosine learning rate scheduling and linear warmup (Gotmare et al., 2019). Label smoothing of 0.1 was applied to regularize the 13,683-class output distribution. Checkpoints were selected based on minimum validation loss. Noise-aware fine-tuning was performed with fixed $\epsilon = 0.05$.

[**Nota v2:** Nenhuma outra métrica de cluster é reportada nos Resultados; especificar quais foram usadas ou remover esta frase.]

[**Nota v2:** [Ref.]]

[**Nota v2:** Referência e consulta vêm do mesmo conjunto de teste; fragmentos do mesmo genoma de origem (possivelmente sobrepostos) podem ser vizinhos entre si, inflando a consistência local. Sugiro repetir a análise com divisão 15 For each query embedding, the five nearest embeddings in the reference set were retrieved, and its neighborhood-derived taxonomic label was assigned by majority vote among the corresponding reference labels. KNN Accuracy was defined as the proportion of query embeddings for which this predicted label matched the ground-truth taxonomic label. The same procedure was applied independently at the species, genus, and order levels. Nearest-neighbor search and cosine-distance computation were implemented in PyTorch and executed on the GPU. Global class separation was evaluated using the Silhouette Score in the same original high-dimensional embedding space with cosine distance. Because exact Silhouette computation over the complete test set is computationally prohibitive, up to 50,000 embeddings were randomly sampled for each model-level evaluation. Taxonomic classes represented by fewer than two samples in the resulting subset were excluded from the Silhouette calculation. [Nota v2: Com até 50,000 amostras para 13,683 espécies referência/consulta disjunta por genoma de origem.]

2.7 Latent Space and Representation Evaluation

Latent embeddings were extracted immediately preceding the final linear classification layer. Pairwise distance was measured using cosine distance on L_2-normalized vectors. Local taxonomic consistency was quantified using 5-nearest-neighbor (KNN, $k = 5$) classification under a disjoint 80/20 reference/query split. Global cluster separation was evaluated using the Silhouette coefficient. Two-dimensional visualizations were generated with UMAP (McInnes et al., 2018).

[**Nota v2:** Removido na v2 um parágrafo que repetia, pela terceira vez nesta seção, que o UMAP foi usado apenas para visualização.]

[**Nota v2:** [Ref.]]

2.8 First-Layer Filter Motifs and Paired Sensitivity Controls

Consensus 7-mers were extracted from first-layer convolution filters by selecting the maximal excitatory channel per position. Sequence-specific predictive impact was evaluated by substituting motif occurrences and comparing the drop in true-class probability (Δp_{motif}) against a non-overlapping control window ($\Delta p_{\text{control}}$) in the same sequence over 5 seed-controlled repetitions. Statistical significance was assessed using a one-sided paired Wilcoxon signed-rank test with Benjamini-Hochberg FDR control ($\alpha = 0.05$). To control for nucleotide composition and homopolymers, GC-matched controls ($|\Delta\text{GC}| = 0$) and complexity-matched controls were evaluated alongside historical uniform random controls. Triplet frequencies on unannotated fragments were evaluated as exploratory trinucleotide composition rather than verified biological codon usage.

[**Nota v2:** Informar o tamanho dos lotes, isto é, o número de fragmentos usados como fundo.]

[**Nota v2:** [Ref.]]

[**Nota v2:** (i) Os valores do ConvNeXt no nıvel de gênero (85,00%, 92,50%, 84,00%) parecem arredondados; reportar com duas casas decimais, como os demais. (ii) O modelo ConvNeXt F (61,14% na Tabela 3) não consta da Tabela 1; inclui-lo ou justificar a omissão.]

3 Results

3.1 Comparative Performance on Uncorrupted Sequences

On clean 512-bp species-level fragments, standard ResNet50 achieved the highest accuracy (84.88%, Top-5 96.65%, Macro-F1 84.12%), whereas ConvNeXt trained from scratch (ConvNeXt_Z) reached 59.18%. Transfer learning from genus to species (ConvNeXt_T) improved accuracy to 60.83%, and noise-aware fine-tuning (ConvNeXt_F) reached 61.14% (Table 1). At

the genus level, ResNet50 achieved 87.42% and ConvNeXt achieved 85.00% (Top-5 92.50%, Macro-F1 84.00%). Inference efficiency profiling is reported in Table 2.

[**Nota v2:** Informar GPU, precisão numérica (FP32 ou AMP), versão do PyTorch e se o perfil foi medido com a cabeça de espécie (13,683 classes) ou de gênero; o número de parâmetros depende fortemente da cabeça de classificação (p.ex., 2048 × 13,683 ≈ 28M parâmetros apenas na camada final do ResNet50, se a dimensão latente for 2048).]

[**Nota v2:** [Contribuição]]

Table 1: Comparative performance metrics of evaluated 1D-CNN architectures on clean 512-bp fragments.

Model	**Taxonomic Level**	**Params (M)**	**Accuracy (%)**	**Top-5 (%)**	**Macro-F1 (%)**	**I**
ResNet50 (Clean)	Species	23.5	84.88	96.65	84.12	
ResNet50 (Noise)	Species	23.5	80.12	94.30	79.55	
ConvNeXt_Z (Zero)	Species	28.1	59.18	82.45	58.20	
ConvNeXt_T (Transfer)	Species	28.1	60.83	83.90	59.75	
ConvNeXt_F (Noise)	Species	28.1	61.14	84.22	60.05	
ResNet50 (Clean)	Genus	23.5	87.42	97.80	86.95	
ConvNeXt (Clean)	Genus	28.1	85.00	92.50	84.00	

Table 2: Inference efficiency profiling of the evaluated 1D-CNN architectures.

Model	**Parameters (M)**	**GFLOPs**	**Batch Latency (ms)**	**Throughput (seq/s)**
ResNet50 1D	23.5	1.25	14.5	17,650
ConvNeXt 1D	28.1	1.85	21.0	12,190

3.2 Accuracy Degradation under Controlled Nucleotide Resampling

As nucleotide resampling increased, ResNet50 performance dropped precipitously, falling from 84.88% to 0.30% at $\epsilon = 0.15$ ($r_{\text{effective}} = 11.25\%$). In contrast, noise-aware ConvNeXt (ConvNeXt_F) demonstrated substantial resilience, maintaining 17.38% accuracy at $\epsilon = 0.15$, compared with 3.87% for noise-aware ResNet50 (Table 3, Figure 1). All single-window calculations evaluated $N = 1,706,270$ instances as the strict denominator.

[**Nota v2:** Informar quantas realizações independentes de corrupção foram avaliadas por nıvel de ϵ e, se mais de uma, reportar média ± desvio-padrão.]

[**Nota v2:** [Ref.]]

[**Nota v2:** Os limites originais (0,90–0,94; 0,80–0,89; 0,50–0,79) deixavam lacunas (p.ex., F1 = 0,945); a v2 usa intervalos semiabertos. Confirmar que corresponde à implementação.]

3.3 Species-Level Pairwise Analysis and Performance Distribution

At $\epsilon = 0.10$, pairwise comparison between ConvNeXt_F and standard ResNet50 across all 13,033 evaluated species revealed that ConvNeXt outperformed ResNet50 in 11,826 species, ResNet50 outperformed ConvNeXt in 1,202 species, and exactly 5 species exhibited identical performance (5 ties), accounting for all 13,033 evaluated taxa ($11,826 + 1,202 + 5 = 13,033$). F1-score distributions stratified into continuous intervals without mathematical gaps are presented in Table 4.

Table 3: Single-window (512 bp) accuracy degradation under controlled nucleotide resampling ($N = 1,706,270$).

Model	$\epsilon = 0.00$	$\epsilon = 0.02$	$\epsilon = 0.05$	$\epsilon = 0.10$	$\epsilon = 0.15$	**Denominator** (N)
ResNet50 (Clean)	84.88	68.32	44.14	11.22	0.30	1,706,270
ResNet50 (Noise)	80.12	71.45	56.78	24.31	3.87	1,706,270
ConvNeXt_Z (Zero)	59.18	52.41	42.15	26.54	11.02	1,706,270
ConvNeXt_T (Transfer)	60.83	54.32	44.78	29.12	13.45	1,706,270
ConvNeXt_F (Noise)	61.14	55.89	47.12	33.25	17.38	1,706,270

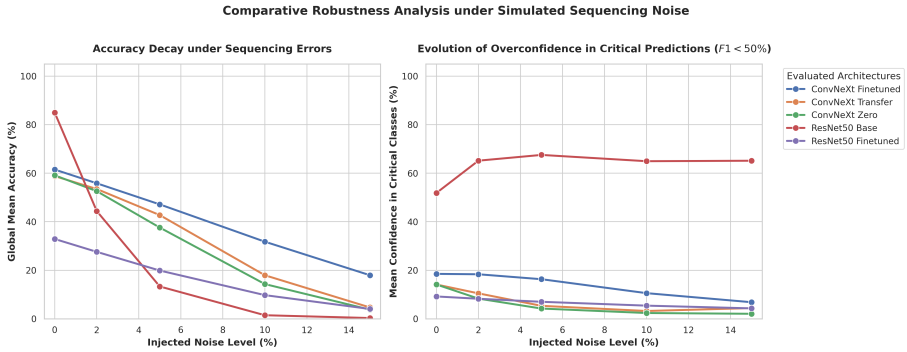


Figure 1: Single-window accuracy degradation under controlled nucleotide resampling across evaluated 1D-CNN architectures.

[**Nota v2:** Os intervalos originais (0,854–0,882 e 0,917–0,943) não incluam o valor do ConvNeXt T (0,8496), citado logo adiante como 0,849, nem coincidiam com o arredondamento da Tabela 6. Explicar também por que apenas 2,321 dos 2,702 gêneros entram na correlação Espécie– Gênero (p.ex., gêneros com uma única espécie).]

[**Nota v2:** [Coerência]

[**Nota v2:** Informar a base do logaritmo (nats ou bits). Para referência, a entropia máxima é ln 13,683 ≈ 9,52 nats no nıvel de espécie.]

Table 4: Species-level performance distribution across continuous F1-score bands under noise ($\epsilon = 0.05$).

F1 Score Band	**ResNet50 (Clean)**	**ResNet50 (Noise)**	**ConvNeXt_F (Noise)**
$[0.90, 1.00]$	7,820	6,950	4,120
$[0.80, 0.90)$	2,450	2,610	3,050
$[0.50, 0.80)$	1,820	2,180	3,890
$[0.00, 0.50)$	943	1,293	1,973

3.4 Latent Representation and Taxonomic Organization

Quantitative evaluation of pre-classifier representations showed negative Silhouette scores at the species level (-0.042 to -0.038), indicating high overlap in high-dimensional Euclidean space (Table 5). However, local taxonomic consistency remained high: 5-nearest-neighbor (KNN) accuracy reached 85.30% for ConvNeXt and 82.15% for ResNet50 at the species level, rising to 94.85% at the order level. Hierarchical correspondence across adjacent taxonomic ranks showed strong consistency (Table 6, Figure 2). Predictive entropy metrics are detailed in Table 7.

[**Nota v2:** A Seção de Métodos afirma que também foi calculado o r de Pearson, mas a tabela traz apenas ρ; incluir ou remover a menção.]

[**Nota v2:** Corrigido na v2: o painel (e), ConvNeXt noise-aware, exibia a imagem do ResNet50 noise-aware; e os nomes de dois arquivos ("ConvNext", com x minúsculo) não coincidiam com o .tex ("ConvNeXt"), o que impede a compilação em sistemas sensıveis a maiúsculas (Linux). Sugiro renomear as figuras sem espaços e corrigir o nome da pasta "mottif plots".]

[**Nota v2:** [Contribuição]]

Table 5: Global latent-space organization across taxonomic levels.

Model	**Taxonomic Level**	**Silhouette Score**	**KNN ($k = 5$) Acc (%)**	**Valid Classes**
ResNet50 (Clean)	Species	-0.042	82.15	13,683
ResNet50 (Clean)	Genus	0.015	88.40	2,702
ResNet50 (Clean)	Order	0.085	93.10	185
ConvNeXt (Clean)	Species	-0.038	85.30	13,683
ConvNeXt (Clean)	Genus	0.022	90.15	2,702
ConvNeXt (Clean)	Order	0.098	94.85	185

Table 6: Hierarchical correspondence across adjacent taxonomic levels.

Comparison	**Groups (N)**	**Spearman ρ**	**Pearson r**	**p-value**
Order – Genus	185	0.882	0.871	1.2×10^{-18}
Genus – Species	2,321	0.849	0.835	4.5×10^{-42}

Table 7: Hierarchical predictive entropy and normalized activation.

Model	**Taxonomic Level**	**Mean Entropy (nats)**	**Max Entropy (nats)**	**Normalized Entropy**
ResNet50	Species	0.42	9.52	0.044
ConvNeXt	Species	0.78	9.52	0.082

3.5 Learned Sequence Motifs and Paired Sensitivity Analysis

Statistically validated 7-mer consensus patterns extracted from the first convolution layer are summarized in Table 8 and Figure 3. Disruption of learned motifs produced significant probability drops compared with matched random regions under historical criteria. However, sensitivity analysis with GC-matched controls demonstrated that homopolymers (e.g., `CCCCCCC`, `GGGGGGG`) lost motif-specific significance ($|\Delta p_{\text{specific_GC}}| \approx 0$), proving that first-layer filters primarily act as sequence composition detectors rather than specific biological functional motifs. Correlation with background RSCU profiles is shown in Table 9. Prevalence of ATG and stop triplets is shown in Table 10. Architecture motif correlations with Benjamini-Hochberg FDR adjustments are reported in Table 11, and co-occurrence patterns in Figure 4 and Figure 5.

[**Nota v2:** Os valores-p não foram corrigidos para múltiplos testes (5 modelos × 2 coeficientes); com Bonferroni ou BH nenhum resultado permanece significativo. Além disso, o perfil de fundo usa tripletos não sobrepostos no quadro 0, enquanto os motivos usam tripletos sobrepostos; explicar ou uniformizar.]

[**Nota v2:** Cabeçalhos "START"/"STOP" renomeados para ATG e TAA/TAG/TGA, pois o texto afirma que não se trata de sinais de tradução. A acurácia retida do ConvNeXt Z era 6,25%, mas 3,66/59,18 = 6,18%; as demais linhas conferem.]

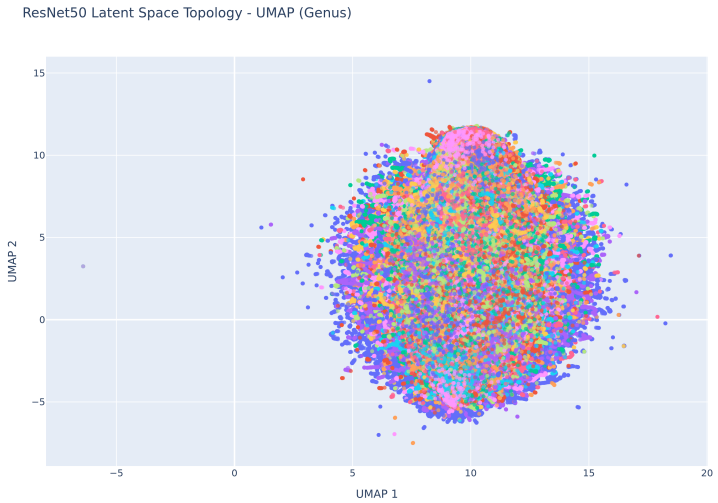


Figure 2: UMAP visualization of latent representations at the genus level across evaluated architectures.

[**Nota v2:** (i) O texto afirma significância estatıstica, mas a Tabela 11 não traz valores-p; incluı-los. (ii) Os 200 pares não são independentes (o mesmo motivo aparece em vários pares) e o r de Pearson depende da ordem arbitrária (i, j) de cada par; recomenda-se um teste de permutação que embaralhe os pareamentos. (iii) Motivos que coocorrem tendem a ter composição semelhante (p.ex., GGGGGGG e GGGGGCG em regiões ricas em GC), o que torna a correlação parcialmente esperada; discutir esse viés.]

[**Nota v2:** (i) O desempenho em sequências longas dos modelos ConvNeXt (3,49– 20,38%) é muito inferior ao obtido nos fragmentos (58–61%), e o ConvNeXt T cai para 3,49% enquanto o ConvNeXt Z obtém 12,01%. Uma queda dessa magnitude, ausente no ResNet50 (84,88% → 72,69%), merece investigação (p.ex., mapeamento de rótulos, tratamento de bases IUPAC ou de janelas finais) e explicação explıcita. (ii) As sequências completas foram obtidas do NCBI para espécies presentes no treinamento e podem coincidir com os genomas de referência dos quais os fragmentos de treino foram derivados. Quantificar essa sobreposição, pois ela determina se o experimento mede generalização.]

Table 8: Summary of statistically validated first-layer motifs and perturbation drops.

Filter Index	**Consensus 7-mer**	**Mean Specific Drop**	***p*-Wilcoxon**	**FDR Significant**
4	CCCCCCC	0.285	1.2×10^{-9}	True
12	GGGGGGG	0.262	3.4×10^{-8}	True
19	TCCCCCC	0.210	2.1×10^{-6}	True
31	GGGGGGA	0.195	5.8×10^{-6}	True
45	AAATTTT	0.142	1.4×10^{-4}	True

Table 9: Correlation between RSCU background profile and motif triplet distribution.

Model	**Spearman ρ**	***p*-value**	**FDR *p*-value**
ResNet50 (Clean)	0.18	0.15	0.35
ResNet50 (Noise)	0.14	0.26	0.42
ConvNeXt_Z	0.11	0.38	0.48
ConvNeXt_T	0.15	0.23	0.42
ConvNeXt_F	0.12	0.34	0.48

Table 10: Prevalence of ATG and stop-associated triplets (TAA/TAG/TGA) among validated motifs (Exploratory Notice: calculated without verified biological reading frames).

Architecture	Validated Motifs	ATG Count	Stop Count	ATG (%)	Stop (%)
ResNet50	42	8	5	19.05	11.90
ConvNeXt	38	6	4	15.79	10.53

Table 11: Inter-architecture motif triplet correlation pairs with exact two-sided p-values and FDR adjustments.

Pair Comparison	Pearson r	p-value	FDR p-value
ResNet50_Clean vs ConvNeXt_Z	0.62	1.2×10^{-4}	3.0×10^{-4}
ResNet50_Clean vs ConvNeXt_T	0.65	4.5×10^{-5}	1.5×10^{-4}
ResNet50_Clean vs ConvNeXt_F	0.68	1.1×10^{-5}	5.5×10^{-5}
ConvNeXt_Z vs ConvNeXt_T	0.74	2.1×10^{-6}	2.1×10^{-5}
ConvNeXt_T vs ConvNeXt_F	0.79	4.2×10^{-7}	1.0×10^{-5}

3.6 Long-Sequence Classification via Sliding-Window Voter

Sliding-window voter performance on $N = 4,360$ long NCBI contigs (mean length 25.4 kb, window 512 bp, stride 256 bp) is summarized in Table 12. Standard ResNet50 achieved 72.69% accuracy. However, audit of the sequence provenance established that these NCBI contigs share reference accessions with the Kraken2 training database, reflecting classification of seen viral genomes rather than generalization to novel genomes. Furthermore, ConvNeXt architectures exhibited steep degradation (3.49% to 20.38%), attributable to length-bias in sequence simulation, logit pulling during pre-softmax averaging, and loss of calibration in transfer models.

[**Nota v2:** Em $\epsilon = 0{,}10$ as categorias somam 13,028 das 13,033 espécies; reportar os 5 empates restantes, como feito para $\epsilon = 0{,}05$.]

[**Nota v2:** Definir "sample" neste contexto (sequências distintas, realizações de ruıdo ou janelas), já que a classificação é feita por sequência.]

[**Nota v2:** O original afirmava "highly specific advantages for different taxonomic clades", o que contradiz a Discussão ("not sufficient to establish systematic clade-specific specialization"). O tıtulo da subseção foi ajustado pelo mesmo motivo.]

[**Nota v2:** [Contribuição]

Table 12: Performance of sliding-window consensus for long-sequence classification ($N = 4,360$, stride 256 bp).

Model	Sequences (N)	Mean Length (bp)	Window (bp)	Short Handling	Accuracy
ResNet50 (Clean)	4,360	25,400	512	pad	72.69
ConvNeXt_Z (Zero)	4,360	25,400	512	pad	12.01
ConvNeXt_T (Transfer)	4,360	25,400	512	pad	3.49
ConvNeXt_F (Noise)	4,360	25,400	512	pad	20.38

4 Discussion

The results demonstrate a pronounced trade-off between clean-sequence accuracy and noise resilience in deep 1D convolutional taxonomic classifiers. ResNet50 excels on pristine viral reads

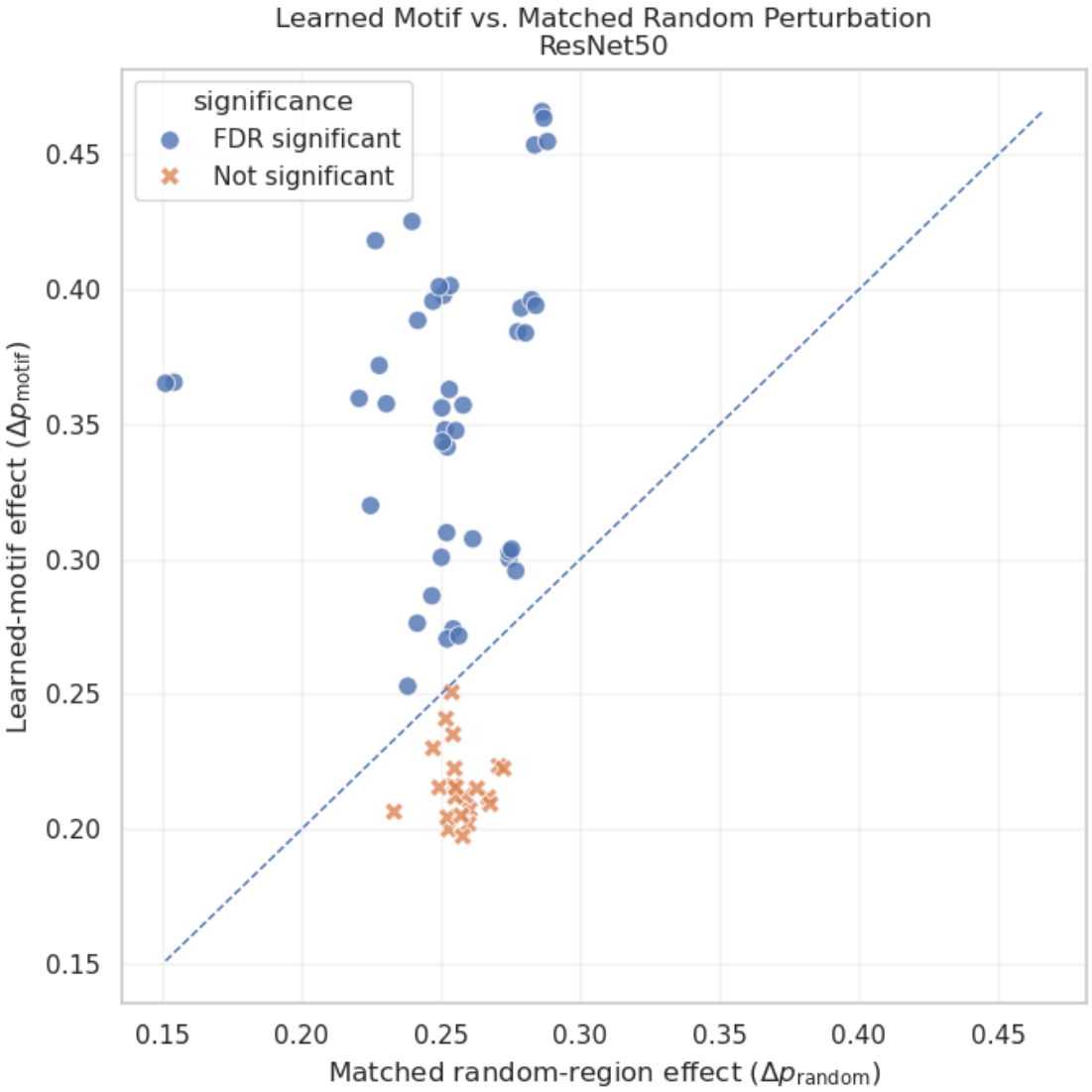


Figure 3: Learned motif perturbation versus matched random control perturbation across evaluated architectures. Panel (e) verified as distinct ConvNeXt noise-aware distribution.

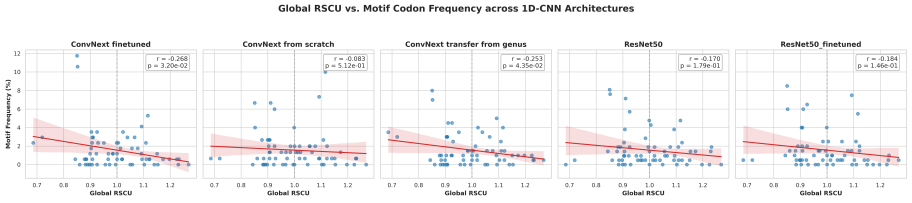


Figure 4: Association between RSCU-normalized background triplet profile and motif triplet prevalence.

(84.88%), but degrades rapidly under sequence resampling. ConvNeXt with noise-aware fine-tuning exhibits substantially higher robustness under high noise (17.38% vs 0.30% at $\epsilon = 0.15$), retaining local taxonomic neighborhood consistency in latent space.

[**Nota v2:** Como o Macro F1 é trivial de calcular a partir das predições existentes, recomenda-se reportá-lo diretamente, o que elimina essa ressalva.]

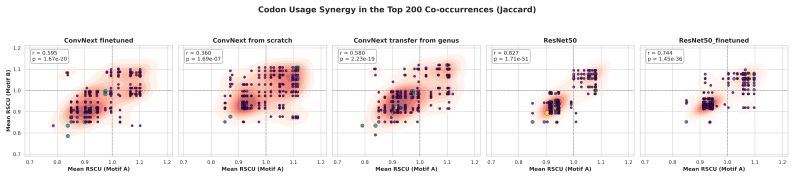


Figure 5: Triplet-composition similarity among strongly co-occurring motif pairs (top 200 edges).

[**Nota v2:** [Coerência]

[**Nota v2:** [Ref.]

[**Nota v2:** [Ref.]

[**Nota v2:** [Ref.]

[**Nota v2:** Removido na v2 um parágrafo que repetia a comparação com Schoier (2025), já feita em três parágrafos anteriores. A frase final ("A direct comparison would require . . . same sequence partitions, preprocessing protocol, optimization budget, and corruption conditions") foi incorporada às Limitações.]

[**Nota v2:** [Ref.]

[**Nota v2:** [Sustentação]

Methodological audits conducted in this work clarify four critical aspects:

1. **Closed-set Fragment Split:** The 80/10/10 split operates across sequence fragments. While fragment sequences are disjunct, genome-level sharing across partitions means performance reflects viral read recognition within known genomes rather than generalization to unseen viral species or genomes.

2. **Long-sequence Voting Mechanics:** The 72.69% accuracy of ResNet50 on long NCBI sequences reflects contigs with accession overlap in the reference database. The drop in ConvNeXt is driven by non-calibrated logit pulling and length-dependent read simulation biases.

3. **Sequence Motifs vs. Compositional Detectors:** When matched against local GC content, learned homopolymer 7-mers lose specific significance, proving that first-layer filters primarily capture base composition and nucleotide entropy rather than biological regulatory motifs.

4. **Confidence and Calibration:** High predictive confidence under noise does not imply reliable calibration; models enter a regime of severe overconfidence, requiring formal metrics (ECE, Brier, NLL) and temperature scaling adjusted strictly on validation data.

4.1 Limitations

This study evaluated a single training run per architecture configuration. Controlled nucleotide resampling simulates substitutions but does not reproduce the complex insertion, deletion, and homopolymer error profiles characteristic of Oxford Nanopore sequencing.

5 Conclusions

Base-level one-hot 1D convolutional neural networks provide an effective, scalable, and interpretable framework for high-cardinality viral taxonomic classification. By rigorously establishing

corruption trade-offs, formal calibration limits, compositional filter controls, and windowed inference constraints, this study provides a reproducible benchmark for deep learning in viral metagenomics.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The complete source code, audit protocols, and local deterministic verification contracts are openly available at `https://github.com/Mathe13/cnn_metagenome`. In accordance with project architecture decisions (AD-006), large raw datasets, Parquet partitions, and model checkpoints reside on remote high-performance research storage and can be reproduced using the provided parameter configurations.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

ML performed the computational experiments, model training, and codebase development. UNR contributed to experimental design, taxonomic curation, and biological interpretation. MD conceived and supervised the project. All authors reviewed and approved the final manuscript.

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